AirFly Insights

Data Visualization and Analysis of Airline Operations

Anand Kumar

Batch:13

Email: anandsibbuchoudhary@gmail.com

Acknowledgement

I would like to express my sincere gratitude to Infosys Springboard for providing me with the opportunity to work on the project titled "AirFly Insights: Data Visualization and Analysis of Airline Operations". This project helped me gain practical knowledge in data analysis, data cleaning, and data visualization using Python.

I would also like to thank all the instructors and mentors who provided guidance and support throughout the project. Their valuable suggestions and learning resources helped me understand the concepts of data preprocessing, exploratory data analysis, delay analysis, and data visualization techniques.

Finally, I would like to thank everyone who directly or indirectly supported me in completing this project successfully. This project was a great learning experience and helped me improve my analytical and technical skills.

Table of Contents

1. Introduction

2. Objective of the Project

3. Dataset Description

4. Tools and Technologies Used

5. Data Cleaning and Preprocessing

6. Exploratory Data Analysis

7. Delay Analysis

8. Route and Airport Analysis

9. Cancellation and Seasonal Analysis

10. Key Insights

11. Conclusion

12. References

Introduction

The airline industry generates a large amount of data every day, including information about flights, delays, cancellations, airports, and routes. Analyzing this data can help airlines and airport authorities understand operational performance and improve their services. Data visualization plays an important role in identifying patterns, trends, and problem areas in airline operations.

The AirFly Insights project focuses on analyzing airline flight data to identify delay patterns, cancellation trends, seasonal effects, and airport performance. This project uses data analysis and visualization techniques to transform raw flight data into meaningful insights. Various charts and graphs are used to understand which airlines have the most delays, which routes are busiest, and what are the main reasons for flight cancellations.

This project demonstrates how data analysis can help in better decision-making in the airline industry by identifying problem areas and improving operational efficiency.

Objective of the Project

The main objective of this project is to analyze airline operations data and identify important patterns and trends related to flight delays, cancellations, and airport performance using data visualization techniques.

The specific objectives of this project are:

- To clean and preprocess the airline dataset
- To analyze flight delays by airline, airport, and time
- To analyze cancellation trends and reasons
- To identify the busiest routes and airports
- To analyze seasonal trends in delays and cancellations
- To create visualizations such as bar charts, line charts, histograms, and heatmaps
- To generate insights that can help improve airline operations and scheduling

Dataset Description

The dataset used in this project is the Airline Flights Dataset obtained from Kaggle. The dataset contains a large number of records related to airline flights, including information about airlines, origin and destination airports, departure and arrival times, delays, cancellation reasons, and flight distances.

Important columns in the dataset include Airline, Origin Airport, Destination Airport, Departure Delay, Arrival Delay, Cancellation Reason, Flight Date, and Distance. The dataset contains both numerical and categorical data, which makes it suitable for data analysis and visualization.

This dataset is very useful for analyzing airline performance, delay patterns, route traffic, and seasonal trends. The dataset was cleaned and preprocessed before performing analysis and visualization.

Tools and Technologies Used

The following tools and technologies were used in this project for data cleaning, analysis, and visualization:

- Python – Programming language used for data analysis
- Pandas – Used for data cleaning and preprocessing
- NumPy – Used for numerical operations
- Matplotlib – Used for basic data visualization
- Seaborn – Used for advanced data visualization
- Plotly – Used for interactive visualizations
- Colab Notebook – Used for writing and running the analysis code
- GitHub – Used for project submission and version control

These tools helped in performing data preprocessing, exploratory data analysis, and visualization efficiently.

Data Cleaning and Preprocessing

In this project, data cleaning and preprocessing were important steps before performing data analysis. The dataset contained missing values, incorrect data types, and unnecessary columns which were handled during preprocessing.

First, the dataset was loaded using Pandas, and basic exploration was performed to understand the structure of the dataset, data types, and missing values. Missing values in delay and cancellation-related columns were handled by either filling them with appropriate values or removing rows where necessary.

New features were created from the existing data to make analysis easier. These features included Month, Day of Week, Hour, and Route (Origin to Destination). Datetime columns were converted into proper datetime format to extract time-based features.

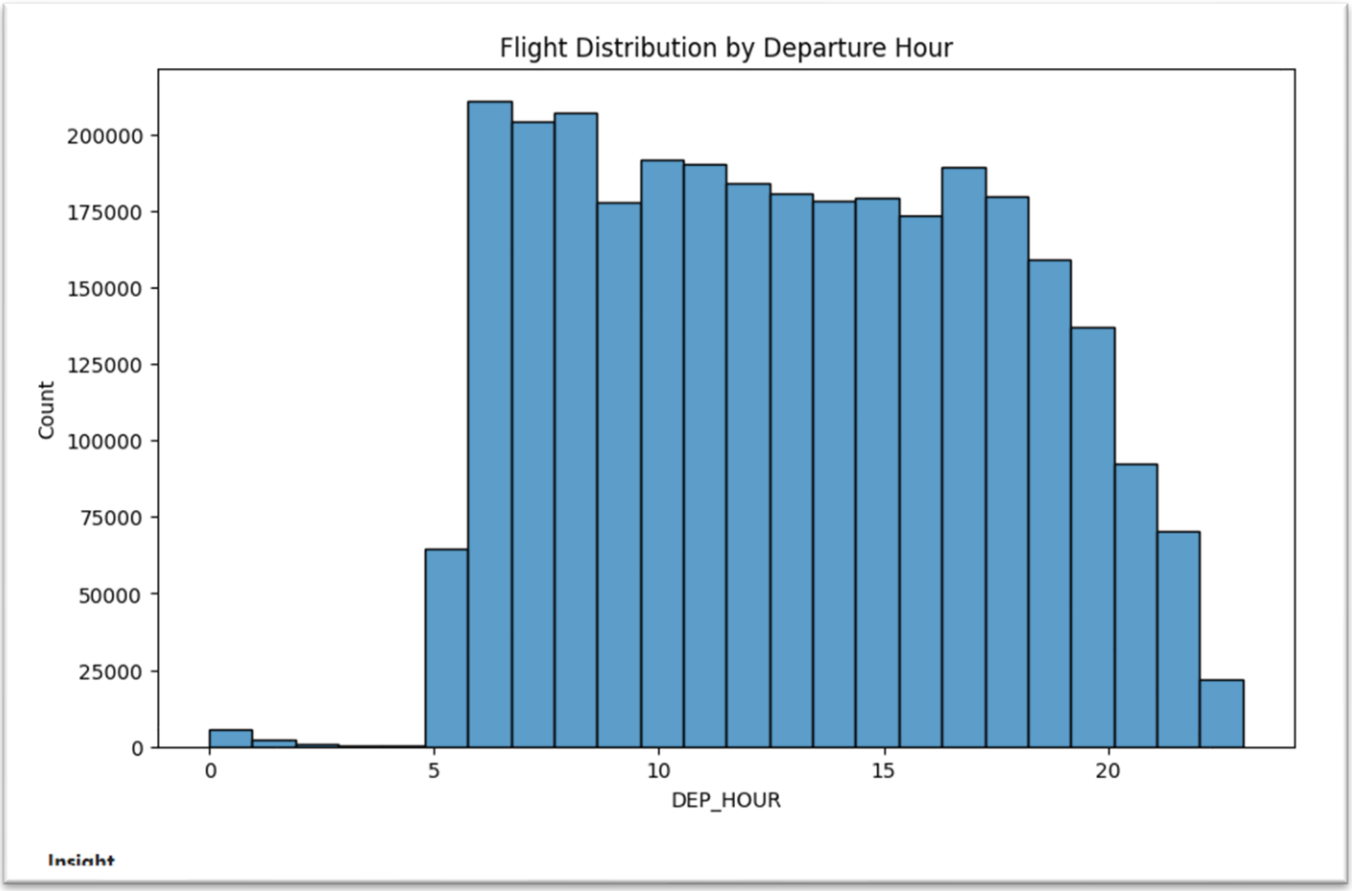
Data types were optimized to reduce memory usage, and a cleaned dataset was saved for further analysis. This preprocessing step made the dataset ready for Exploratory Data Analysis and visualization.

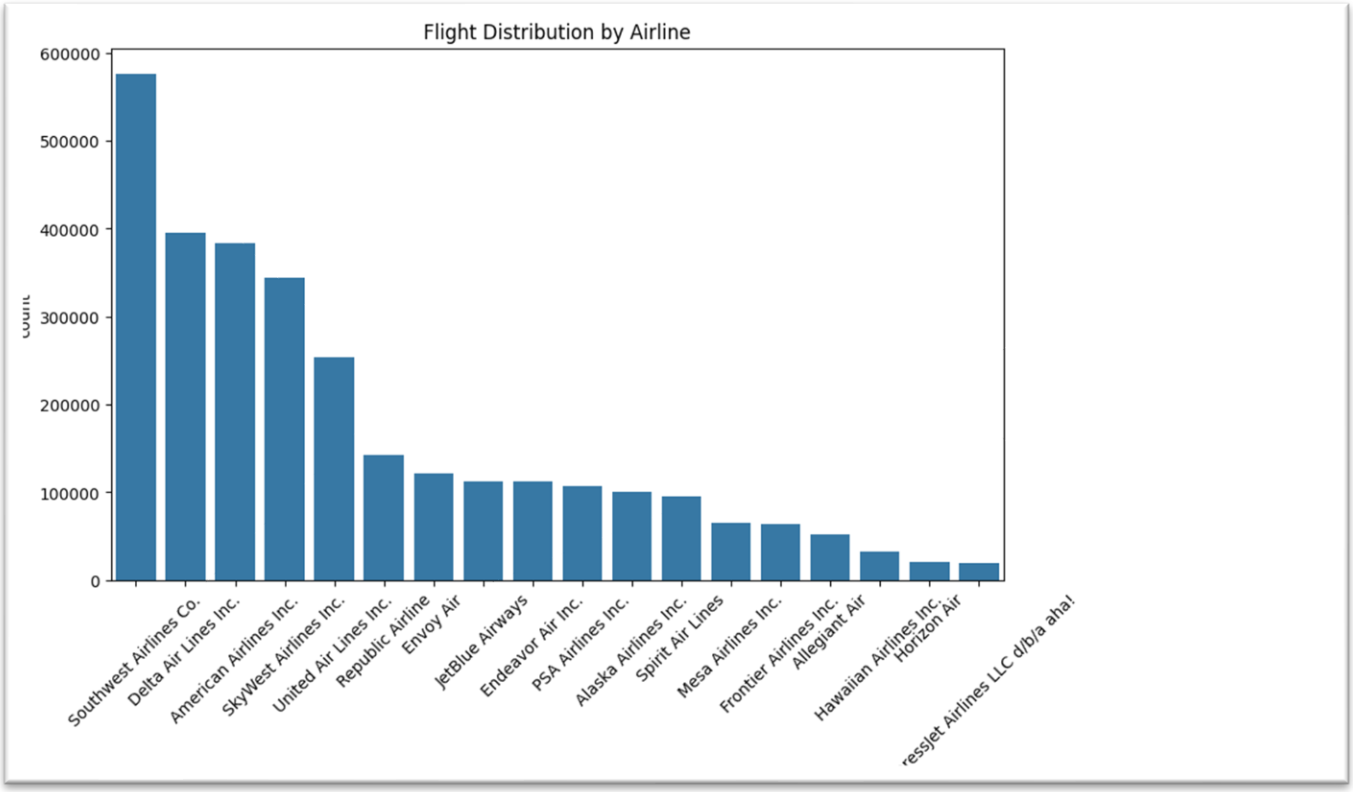
Exploratory Data Analysis

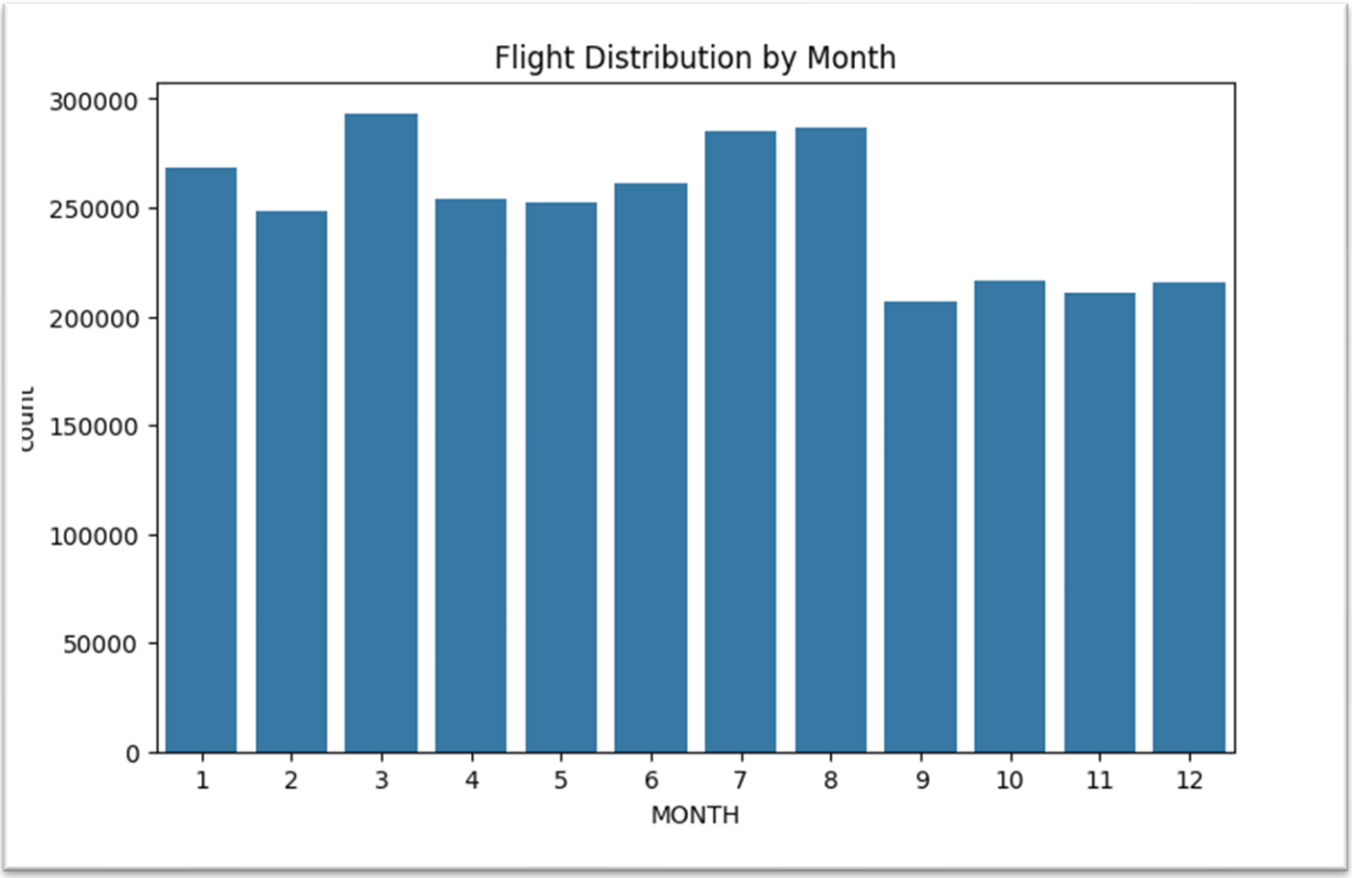
Exploratory Data Analysis was performed to understand the patterns and distribution of the data. Various visualizations were created to analyze airline performance, airport traffic, and flight distribution.

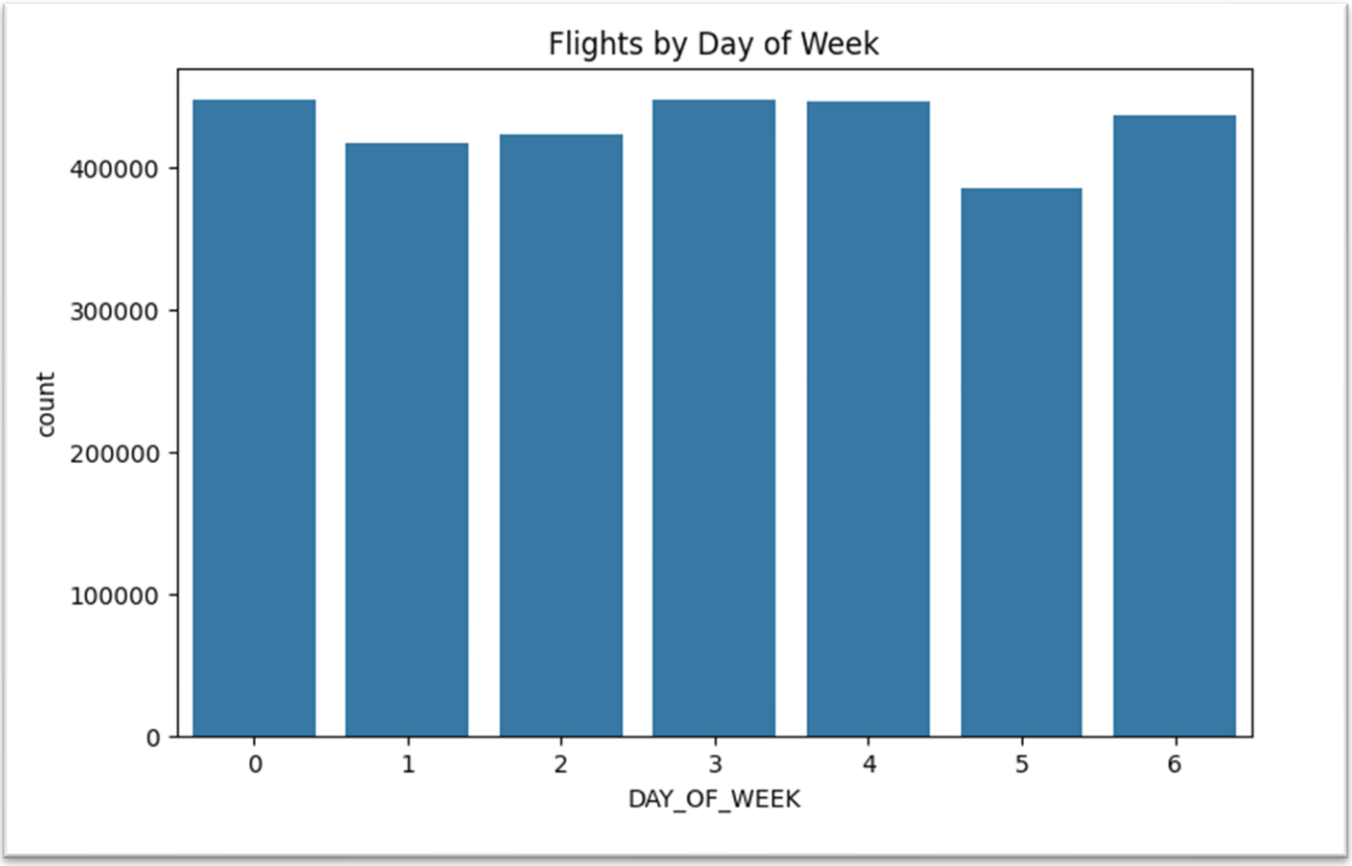
Bar charts were used to identify the top airlines with the highest number of flights and the busiest airports. Line charts were used to analyze monthly trends and seasonal patterns. Histograms and boxplots were used to understand the distribution of delays and identify outliers.

The analysis showed that some airports handle a higher number of flights compared to others, and certain months have higher flight traffic. These visualizations helped in understanding the overall structure of the dataset and identifying important trends.


Flight Distribution by Departure Hour
Count
200000
175000
150000
125000
100000
75000
50000
25000
0
0
5
10
15
20
DEP_HOUR
Insight


Flight Distribution by Airline
600000
500000
400000
300000
200000
100000
0
Southwest Airlines Co.
Delta Air Lines Inc.
American Airlines Inc.
SkyWest Airlines Inc.
United Air Lines Inc.
Republic Airline
Envoy Air
JetBlue Airways
Endeavor Air Inc.
PSA Airlines Inc.
Alaska Airlines Inc.
Spirit Air Lines
Mesa Airlines Inc.
Frontier Airlines Inc.
Allegiant Air
Hawaiian Airlines Inc.
Horizon Air


Flight Distribution by Month
300000
250000
200000
150000
100000
50000
0
count
1
2
3
4
5
6
7
8
9
10
11
12
MONTH


Flights by Day of Week
400000
300000
200000
100000
0
count
0
1
2
3
4
5
6
DAY_OF_WEEK

Delay Analysis

Delay analysis was performed to identify the main causes of flight delays and the airlines with the highest delays. Different types of delays were analyzed, including carrier delays, weather delays, NAS delays, and security delays.

Bar charts and line charts were used to compare delays across different airlines and airports.

The analysis showed that carrier delays and weather delays are the most common reasons for flight delays. It was also observed that delays are higher during evening hours due to air traffic congestion.

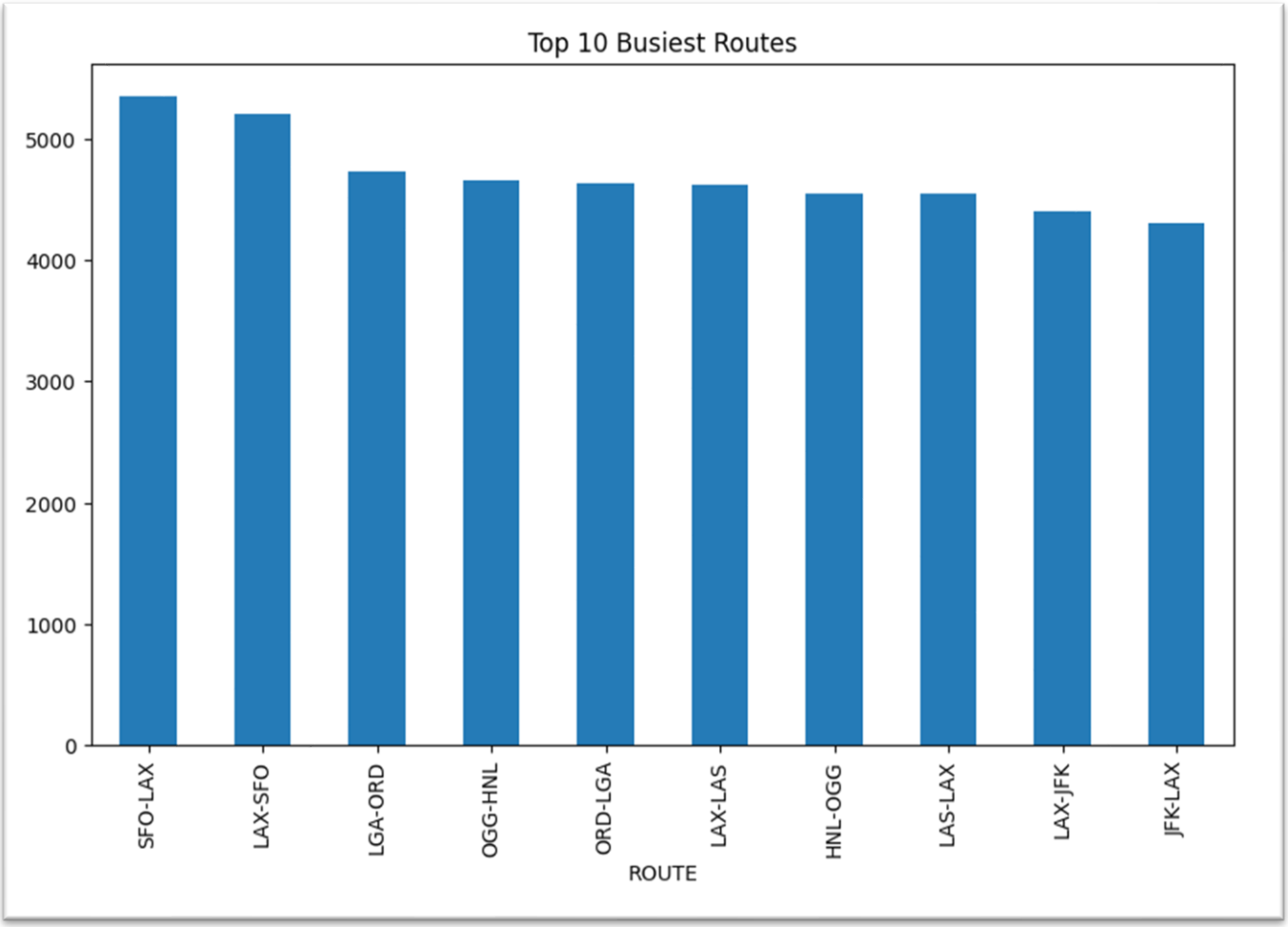
This analysis helps in understanding which factors contribute most to delays and how airlines and airports can take corrective actions to reduce delays.

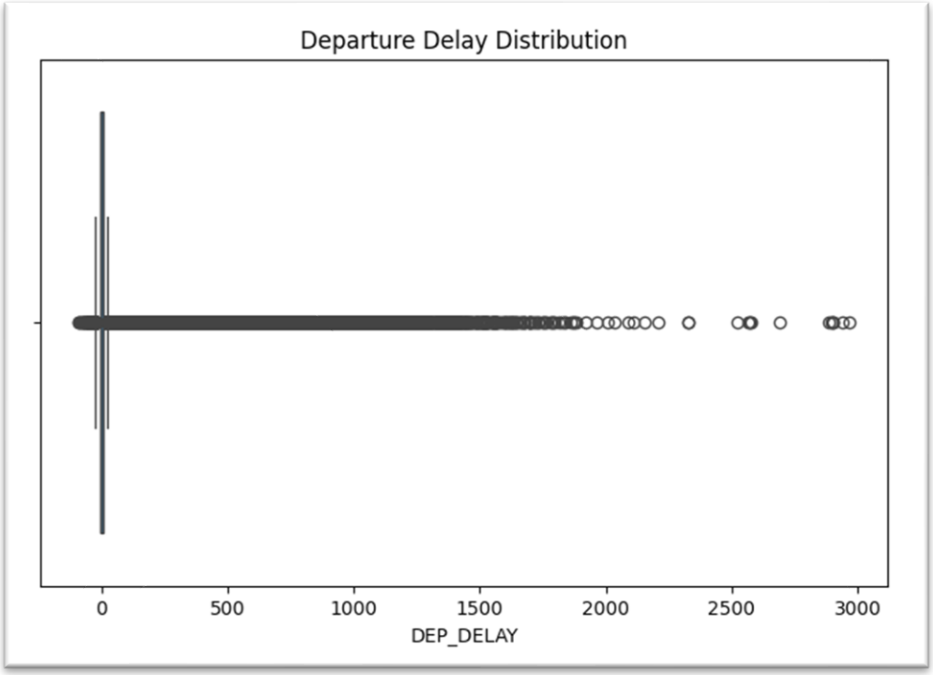
Route and Airport Analysis

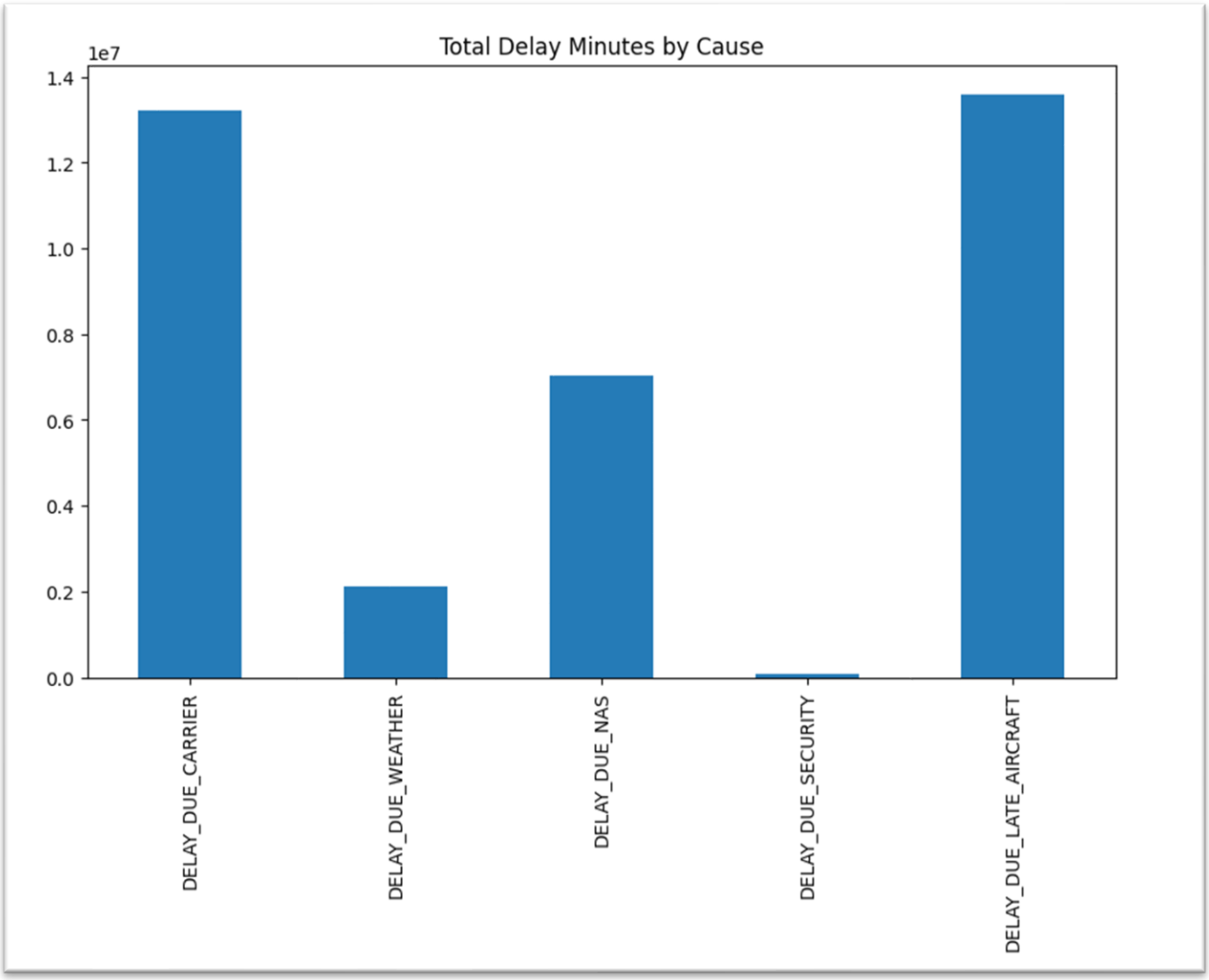
Route and airport analysis was performed to identify the busiest routes and airports and to analyze delay patterns at different airports. The top origin-destination routes were identified based on the number of flights. These routes represent the most frequently used flight paths.

Heatmaps were created to visualize delays across different airports and routes. The analysis showed that some airports experience higher delays due to heavy traffic and congestion. Busiest airports were identified based on the number of departures and arrivals.

This analysis helps airlines and airport authorities understand traffic patterns and manage flight schedules more efficiently to reduce congestion and delays.


Top 10 Busiest Routes
5000
4000
3000
2000
1000
0
SFO-LAX
LAX-SFO
LGA-ORD
OGG-HNL
ORD-LGA
LAX-LAS
HNL-OGG
LAS-LAX
LAX-JFK
JFK-LAX
ROUTE


Departure Delay Distribution
0
500
1000
1500
2000
2500
3000
DEP_DELAY

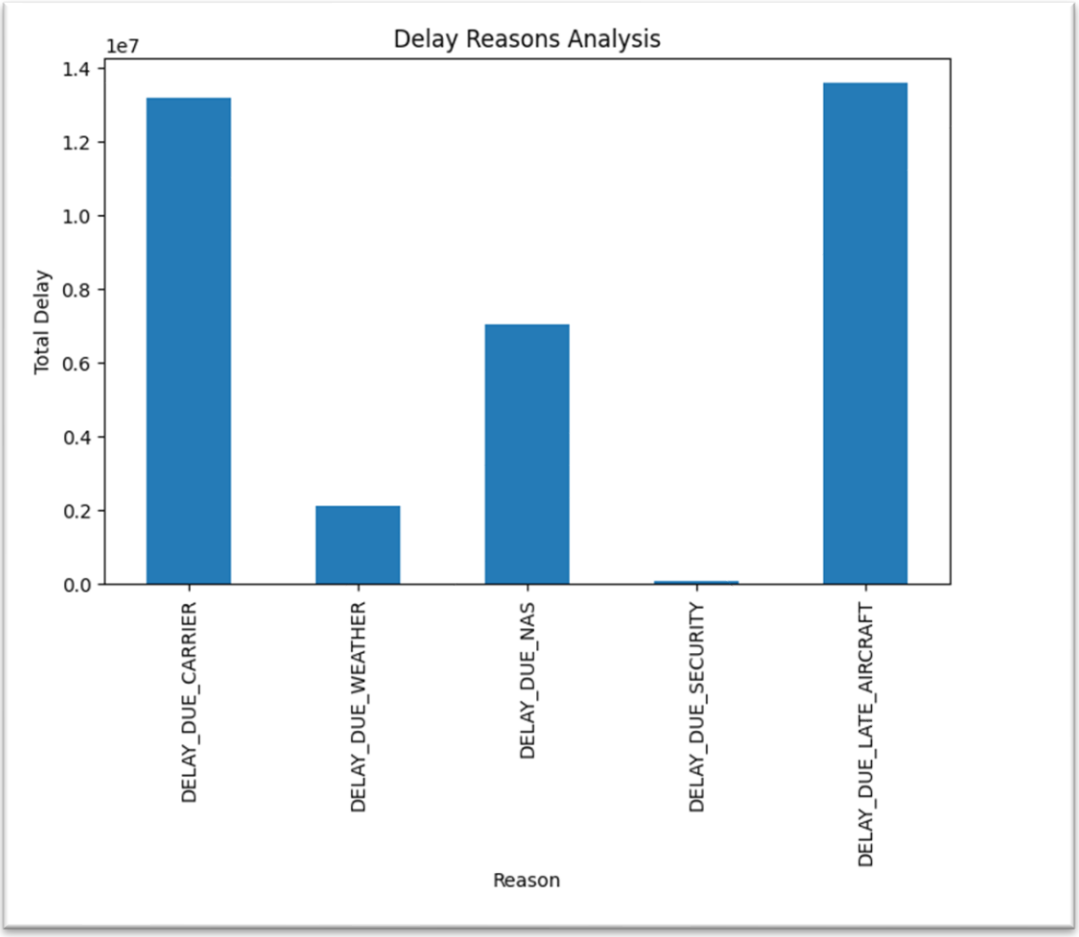


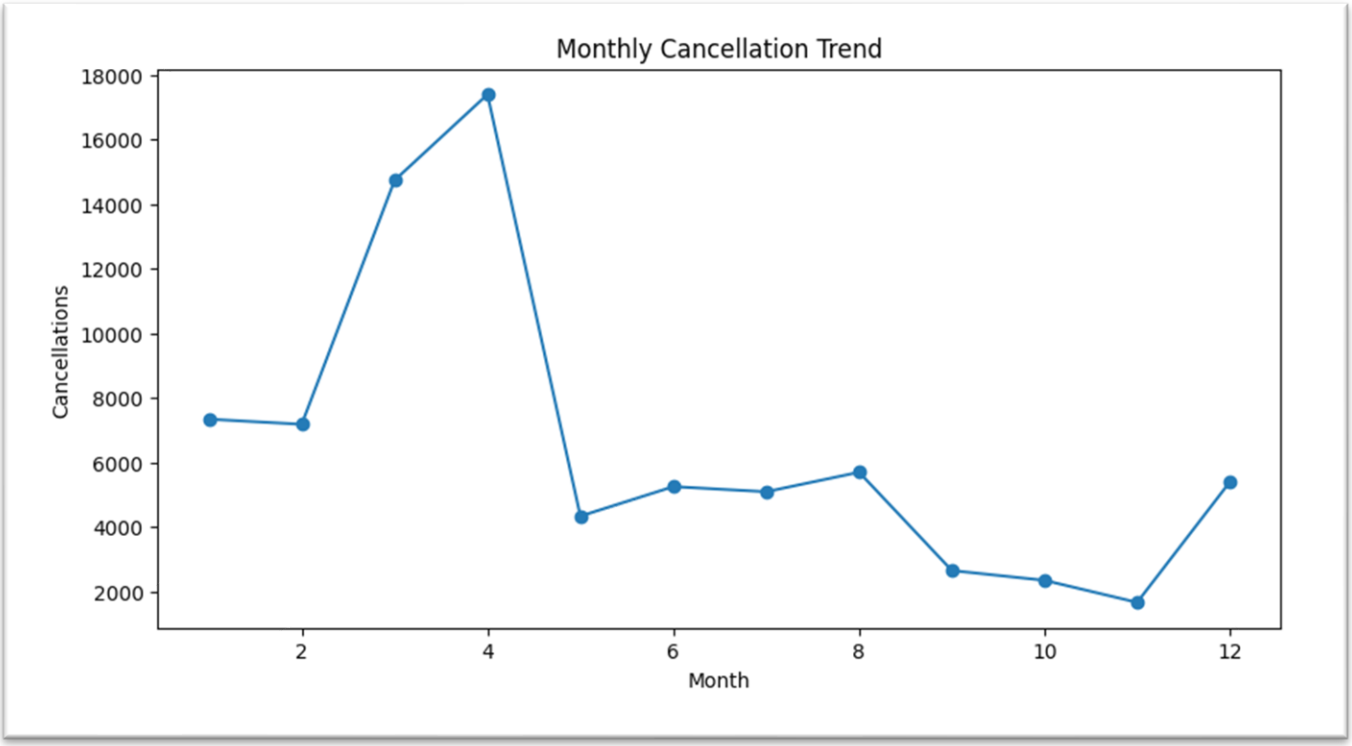
Cancellation and Seasonal Analysis

Cancellation analysis was performed to understand the reasons for flight cancellations and monthly cancellation trends. The main cancellation reasons include carrier issues, weather conditions, NAS (National Aviation System), and security reasons.

Line charts were used to visualize monthly cancellation trends, which showed that cancellations are higher during winter months due to bad weather conditions. Holiday seasons also show higher flight traffic, which sometimes leads to more delays and cancellations.

Seasonal analysis helps in understanding how different times of the year affect airline operations and how airlines can prepare better for peak seasons and bad weather conditions.


Delay Reasons Analysis
1e7
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Total Delay
DELAY_DUE_CARRIER
DELAY_DUE_WEATHER
DELAY_DUE_NAS
DELAY_DUE_SECURITY
DELAY_DUE_LATE_AIRCRAFT
Reason


Monthly Cancellation Trend
18000
16000
14000
12000
10000
8000
6000
4000
2000
Cancellations
2
4
6
8
10
12
Month

Key Insights

The analysis of airline data revealed several important insights:

- Carrier delays and weather delays are the main reasons for flight delays.
- Some airports and routes are more delay-prone due to heavy traffic.
- Evening flights have a higher chance of delays compared to morning flights.
- Winter months have higher flight cancellations due to weather conditions.
- The busiest airports handle a large number of flights, which can lead to congestion and delays.
- Proper scheduling and planning can help reduce delays and cancellations.

These insights can help airlines and airport authorities improve their operations and reduce delays.

Conclusion

The AirFly Insights project analyzed airline operations data to identify patterns in delays, cancellations, and airport performance. The project involved data cleaning, preprocessing, exploratory data analysis, and data visualization.

The analysis showed that delays are mainly caused by carrier and weather issues, and cancellations are more common during winter months. Some airports and routes experience more delays due to heavy traffic and congestion.

This project demonstrates how data visualization and analysis can help in understanding airline operations and improving decision-making. In the future, this project can be extended by creating an interactive dashboard using Streamlit or Power BI for real-time airline data analysis.

References

- Kaggle Airline Flights Dataset
- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation
- Plotly Documentation
- Infosys Springboard Learning Platform

Thank You!

Priya

Mentor

Infosys
Springboard